INCH-POUND

MIL-DTL-13466F
21 August 2025
SUPERSEDING
MIL-B-13466E
16 September 1988

DETAIL SPECIFICATION

BRAID, TEXTILE AND LACE, VELLUM, WOVEN, TEXTILE

(FOR DRESS UNIFORMS)

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This document covers textile braids and woven laces for dress uniforms worn by military personnel of the DoD.

1.2 Classification. The braids and laces will be of the following types, classes, and styles as specified (see 6.2):

Type I - Braid

Class 1 - Soutache
Class 2 - Cord-edge
Class 3 - Combination
Class 4 - Plain with vellum stripe

Type II - Lace, Woven

Class 1 - Plain with vellum stripe

Style A - 1/2-inch
Style B - 1-1/2 inch
Style C - 3/4-inch
Style D - 1-inch

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Type II - Lace, Woven (cont'd)

Class 2 - Colored center stripe

Style A - 3/4-inch
Style B - 1-3/4 inch

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52094 - Thread, Cotton

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1487 - Glossary of Cloth Coating Imperfections
MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS

Title 16 CFR Part 303 - Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ecfr.gov)

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP8	- Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9	- Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM15	- Test Method for Colorfastness to Perspiration
AATCC TM16.3	- Test Method for Colorfastness to Light: Xenon-Arc
AATCC TM117	- Test Method for Colorfastness to Heat: Dry (Excluding Pressing)
AATCC TM132	- Test Method for Colorfastness to Drycleaning
AATCC TM135	- Test Method for Dimensional Changes of Fabrics after Home Laundering
AATCC TM158	- Test Method for Dimensional Changes on Drycleaning, Perchloroethylene: Machine

(Copies of these documents are available online at https://www.aatcc.org)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 – Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org)

ASTM INTERNATIONAL

ASTM D3775	- Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3990	- Standard Terminology Relating to Fabric Defects
ASTM D5035	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D6193	- Standard Practice for Stiches and Seams

(Copies of these documents are available online at www.astm.org)

OTHER PUBLICATIONS

SDL ATLAS (for cloth)

Part Number 402985 – Slub/Knot Replica Set

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(Replica set is available for purchase from SDL Atlas Customer Service, 1-803-329-2110)

2.4 Order of precedence. Unless otherwise noted here or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection in accordance with 4.3.1.

3.2 Standard sample. Unless otherwise indicated, the finished braid/lace shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Basic materials.

3.4.1 Cotton yarn. The yarns shall be cotton in the sizes specified in Table I and 3.6.

3.4.2 Rayon or nylon yarn. The yarns shall be multifilament bright or semi-dull rayon cuprammonium or viscose or nylon (singles) of a suitable denier to meet the requirements of this document as specified in Table I and 3.6.

3.4.3 Mohair yarn. The yarns shall be not lower in grade than 28's and shall be carded, combed, and spun on the worsted system and twisted into 2/30 (± 1 count) yarn as specified in Table I and 3.6.

3.4.4 Polyester yarn. The yarns shall be spun, textured, or filament polyester in the sizes specified in Table I and 3.6.

3.4.5 Weave for Type II, Class 1, Style A lace. The weave shall be a filling effect weaving as follows:

Every pick over and under - 4 ends
Every pick over and under - 4 ends
Every pick over and under - 24 ends
Every pick over and under - 2 ends
Every pick over and under - 24 ends

The center warp section of 4 ends 3 times shall be woven as follows:

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First 4 ends over and under every pick
Second 4 ends over 1 pick - under 1 pick
Over 1 pick - under 2 picks
Over 1 pick - under 1 pick, over 1 pick
Third 4 ends over and under every pick

Every pick over and under - 24 ends
Every pick over and under - 2 ends
Every pick over and under - 24 ends
Every pick over and under - 4 ends
Every pick over and under - 4 ends

3.4.5.1 Vellums. The vellums on the lace shall be symmetrical with both sets of vellums the same width and equidistance from the edges of the lace.

3.4.6 Weave for Type I, Class 3 polyester braid. The weave shall be as follows:

Eight ends to form a square of the ground warp, even number ends up for four picks, odd number ends down for the same four picks.

The second eight ends weave the same as the first eight ends except that they start after the first two picks.

Repeat this weave pattern across entire width (a total of 128 ends full width). Selvage weaves plain, one up, one down and is held at the end by a catch-cord using latch needle construction and shall contain thirty-two picks per inch minimum.

TABLE I. Components and materials.

Type	Class	Component material	Characteristics	Requirement Reference
I II	1, 2, 3, 4 1, 2	Rayon or nylon yarn	Material, luster, filament and ply	3.6.1
I	1	Cotton yarn	Material, count, ply	3.6.1
I	2	Cotton core and stuffer yarn Cotton flat braid yarn	Material, count, ply Material, count, ply	3.6.2.1.1 3.6.2.2
I	3 (1/2-inch)	Cotton stuffer yarn	Material, count, ply	3.6.3.1
I	3 (1-1/4 inch)	Mohair yarn	Material, grade, processes, size	3.6.3.2
I	3 (1-1/4 inch)	Polyester yarn	Material, yarn size, ply	3.6.3.2
II	1, 2	Warp and stuffer yarn	Material, mercerized, count, ply	3.6.5

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3.5 Color. The color(s) of the dyed braids and laces shall be as specified in Table II. The yarns for the braids and woven laces shall be yarn dyed. The nylon and polyester yarns may be spinning solution dyed. The finished item shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced.

TABLE II. Color requirements.

Component	Type	Class	Color 1/
Braid, soutache	I	1	Goldenlite, Cable No. 67107
Braid, cord-edge:			
Cord portion	I	2	Goldenlite, Cable No. 67107
Flat braid portion	I	2	Black, Shade No. 65018
Braid, combination:			
1/2-inch	I	3	Goldenlite, Cable No. 67107
1-1/4 inch	I	3	Black, Shade No. 65018
Braid, plain	I	4	Goldenlite, Cable No. 67107
Lace, plain	II	1	Goldenlite, Cable No. 67107
Lace, center stripe	II	2	2/

1/ Cable numbers are from the Department of Defense Standard Color Card for Heraldic Yarns, Embroidery, and Ribbons.

2/ The edge stripes shall be Goldenlite, Cable No. 67107, and the center stripe(s) color(s) shall be as specified in 3.6.5.1.

3.5.1 Colors for Type II, Class 2 lace. The colors shall be as specified in Table III and 3.6.5.1.

TABLE III. Colors for Type II, Class 2 Lace.

Arms and Services	Color	Cable No. 1/
Adjutant General's Corp	Dark Blue or Scarlet	65012 65006
Armor Artillery Aviation	Yellow Scarlet Ultramarine or Golden Orange	65002 65006 65010 65003
Chaplains Chemical Corps	Black or Cobalt Blue	65018 65011
Corps of Engineers	Scarlet or White	65006 65005
Finance Corps	Silver Gray or Golden Yellow	65008 65001

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TABLE III. Colors for Type II, Class 2 Lace. - Continued

Arms and Services	Color	Cable No. 1/
Infantry Inspector General	Light Blue Dark Blue or Light Blue	65014 65012 65014
Judge Advocate General's Corp	Dark Blue or White	65012 65005
Army Medical Service Corps	Maroon or White	65017 65005
Intelligence and Security	Oriental Blue or Silver Gray	65027 65008
Military Police Corps	Green or Yellow	65007 65002
National Guard Bureau Ordnance Corps	Dark Blue Crimson or Yellow	65012 65013 65002
Permanent Professors of the U.S. Military Academy Quartermaster Signal Corps	Scarlet Silvery Gray Buff Orange or White	65006 65008 65015 65004 65005
Staff Specialists Reserve Transportation Corps	Green Brick Red or Golden Yellow	65007 65020 65001
Unassigned to Branch	Teal Blue or White	65024 65005
Warrant Officers	Brown	65016

1/ Cable numbers in the 65,000 series are from the DoD Standard Shades for Heraldic Yarns.

3.5.2 Visual shade matching. The color and appearance of the dyed braid shall match the standard sample (see 6.2) when tested as specified in 4.4.

3.5.3 Labile sulfur. The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid is prohibited. The yarns shall be "free" of labile sulfur when tested as specified in 4.6.2.

3.5.4 Colorfastness (all classes). Unless otherwise specified in the contract or procurement documents, the dyed and finished braids and woven laces specified in 3.3.1 through 3.3.4 shall conform to the colorfastness requirements listed below in Table IV when tested as specified in 4.5.

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TABLE IV. Colorfastness requirements (All Classes).

Colors	Laundering Color Change and Staining, grade, (3 cycles), (min.)	Light (after 40 AFU or 170kJ/ (m^2nm)@420 nm) grade, (min.) 1/	Perspiration Color Change and Staining, grade, (min.)	Crocking (wet and dry) grade, (min.)	Drycleaning, Color Change and Staining, rating (min.)
All Classes	4.0	4.0	4.0	3.0	3.5

1/ AFU: AATCC Fading Units

3.6 Construction.

3.6.1 Type I, Class 1. The soutache cover shall be made of rayon or nylon yarns braided around two cotton warp ends (see 3.4.1 and 3.4.2). Each cotton warp end shall be a 20/3/4-ply cord or equal. The soutache shall conform to the physical requirements in Table V.

3.6.2 Type I, Class 2. The flat braid section and the cord shall be braided together in one operation on a cord-edge braider.

3.6.2.1 Cord cover. The cord cover shall be composed of rayon or nylon yarns (see 3.4.2) braided around a braided core (see 3.6.2.1.1) and shall conform to the physical requirements in Table V.

3.6.2.1.1 Core. The core cover shall be braided with a maximum of 38/2 cotton yarn or equivalent around a cotton or textured polyester of appropriate size and construction must be used as a stuffer to give form to the cord section of the cordedge. The yarn for the stuffer shall be a maximum of 10/1 and shall match the color of the cover yarns. The core shall conform to the physical requirements in Table V.

3.6.2.2 Flat braid. The flat braid section shall be braided with 28/2 to 32/2 cotton yarn, dyed Blue 173 and shall conform to the physical requirements in Table V.

3.6.3 Type I, Class 3. The 1/2-inch braid or lace and the 1-1/4 inch braid shall be stitched together using Stitch Type 304 and Seam Type FSa-1 conforming to ASTM D6193 using thread type specified in 3.7. The seam shall be stitched with 10 to 12 stitches per inch.

3.6.3.1 Braid, 1/2-inch. The braid or lace shall be nylon or rayon (see 3.4.2) containing a braided cotton stuffer for each vellum and one cotton stuffer end at each edge of the braid. The cotton yarn for the stuffers shall be 30/2 count. The 1/2-inch braid shall conform to the physical requirements in Table V or Table VI.

3.6.3.2 Braid, 1-1/4 inch. The braid shall be composed of cotton and mohair yarn (see 3.4.1 and 3.4.3) 40/2 (± 1 count). As an alternate, the braid shall be composed of polyester yarn

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(see 3.4.4) 150/2 denier warp, 20/2 count filling and 70 denier catch-cord (nominal yarn size).

3.6.4 Type I, Class 4. The braid shall be rayon or nylon (see 3.4.2) containing a braided cotton stuffer for each vellum and one cotton stuffer end at each edge of the braid. The 9/32-inch braid shall conform to the physical requirements in Table V.

3.6.5 Type II, Classes 1 and 2. All warp and stuffer yarns shall be 38/2 to 40/2 mercerized cotton except that the center stripe yarn for Class 2, Styles A, B, C and D shall be rayon or nylon (see 3.4.2) as an alternate polyester may be used for the filling yarn. The filling yarn for all laces shall be rayon, nylon or polyester (see 3.4.2 and 3.4.4). The laces shall conform to the physical requirements in Table VI.

3.6.5.1 Colored center stripe. The colored center stripe shall be in accordance with the cable number for the branch of service as designated (see 3.5.4). The center stripe shall match the first color designated for branch of service except that when two colors are designated for Class 2, Style A and B the 3/4-inch center stripe shall have a 1/8-inch piping of the second named color on each side of a 1/2-inch stripe of the first named color.

3.7 Thread. The cotton thread shall be dyed Blue AB, C.A. 66044 and shall conform to Type I, Tex 50, 3-ply of A-A-52094.

3.8 Braid and lace requirements. The braid and lace physical requirements shall be as specified in Tables V and VI, respectively.

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TABLE V. Physical requirements for Type I braid.

Class	Width (inches) Min.	Weight per gross yard (ounces)	Number of carriers Min.	Number of carriers Max.	Ends per carrier Min.	Ends per carrier Max.	Number of vellum	Stuffer Yarns Min.	Stuffer Yarns Max.	Warp Ends Min.	Warp Ends Max.	Lines Min.	Lines Max.	Picks per inch Min.	Breaking strength (pounds) Min.	Type braid
1	1/8	8.5	13	-	1	-	-	-	-	2	-	-	-	32	-	Soutache
2	3/32	24.0	19	-	3	-	-	-	-	-	-	-	-	23	-	Plain braid over 2/under 2
	Core	-	8	12	2	3	-	28	32	-	-	-	-	24	-	Plain braid over 2/under 2
	Flat Braid 3/16 1/2 1/	-	21	25	1	1	-	-	-	10	12	5	6	26	-	Plain braid over 2/under 2
3		20.0	49	-	2	-	2 (braided core)	-	-	2	-	-	-	30	-	2/32-inch plain braid over 1 cotton end 3/32-inch vellum 6/32-inch plain braid 3/32-inch vellum 2/32-inch plain braid over one cotton end

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TABLE V. Physical requirements for Type I braid. – Continued

Class	Width (inches) Min.	Weight per gross yard (ounces)	Number of carriers Min. Max.	Ends per carrier Min. Max.	Number of vellum	Stuffer Yarns Min. Max.	Warp Ends Min. Max.	Lines Min. Max.	Picks per inch Min.	Breaking strength (pounds) Min.	Type braid
3	1-1/4	72	58 - braiding in pairs	3 -	-	- 28 spools of 2 ends	- -	16 -	8	125	Basket or diamond Braid over 6/under 6 cotton end 2/32-inch See 3.4.7
3	1-1/4	38 2/	- -	- -	-	- -	128 2/	- -	32 2/	300	1/32-inch plain braid over cotton end 2/3-inch vellum 3/32-inch plain 2/32-inch vellum 1/32-inch plain braid over 1 cotton end
4	9/32	15	32 -	1 -	2 (braided core)	- -	2 -	- -	34	-	

1/ Braided on a bi-circle or radial braider.
2/ Applies to polyester braid woven on a shuttleless loom.

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TABLE VI. Physical requirements for Type II woven laces.

Class	Style	Width (+1/32), inches	Total no. of ends (warp), Min.	Weight per lin. yard (oz.), Min.	Picks per inch (fill), Min.	Number of vellum	Weave pattern	Ribs per inch (center stripe)	Warp breaking strength (lbs.)
1	A	1/2	128	0.20	230 1/	2	Filling effect (see 3.4.5)	-	110
1	B	1-1/2	300	0.50	230	2	Filling effect	-	160
1	C	3/4	170	0.24	230	2	Filling effect	-	120
1	D	1	230	0.30	230	2	Filling effect	-	130
2	A	3/4	172 2/	0.24	230	-	Filling effect 3/	22	120
2	B	1-3/4	456 4/	0.60	230	-	Filling effect 5/	22	200

1/ Two picks per shed, when using needle or shuttless loom.
2/ Arranged 51 left; 70 center stripe; 51 right.
3/ 1/4-inch gold color, 1/4-inch branch of service color, 1/4-inch gold color.
4/ Arranged 128 left; 200 center stripe; 128 right.
5/ 1/2-inch gold color, 3/4-inch branch of service color (see 3.6.5.1), 1/2-inch gold color.

3.8.1 Finish. The Type II, Class 1 lace shall be preshrunk. The lace may have a finish treatment applied to assist in stabilizing the lace and preventing waviness and puckering. Any finish shall meet the toxicity requirements in 3.13.

3.8.2 Dimensional change to dry-cleaning. The Type II, Class 1 lace shall not shrink or elongate more than one (1) percent in the warp direction after one (1) dry-cleaning when tested as specified in 4.4.4.

3.8.3 Dimensional change to laundering. The Type II, Class 1 lace, when manufactured with a polyester fill, shall not shrink or elongate more than one (1) percent in the warp direction after 3 complete laundering cycles when tested as specified in 4.4.4.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the braid or lace of one class, style, and color only, shall be put up on commercial size double-headed spools and flat clipboards as follows:

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a. Type I, Classes 1, 2 and 4 shall be wound on a chipboard spool consisting of a tube with mechanically attached circular ends. The total amount wound on each spool shall be 144 (± 10) yards.

b. Type I, Class 3 and Type II, Classes 1 and 2 shall be wound on a flat chipboard board having a minimum thickness of 0.050 inch and measuring approximately 10-inches in length by 6-1/4 inches in width the total amount wound on each shall be 20 (± 2) yards. Alternatively, a pancake for wide goods with the outer diameter measuring between 11 and 12-inches.

There shall be not more than three lengths per spool for 144-yard spools, and not more than two lengths per board. No one length shall be less than 5 yards.

3.10 Marking. Each spool or board shall have firmly attached a white paper label printed on black with the following information:

STOCK NUMBER
ITEM DESCRIPTION
DOCUMENT NUMBER
CONTRACTOR'S NAME
CONTRACT NUMBER AND DATE
NAME OF CONTRACTING AGENCY
GROSS YARDS

3.11 Fiber identification. Each roll of finished braid shall be labeled or ticketed for fiber content in accordance with title 16 CFR Part 303 Rules and Regulations under the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished braid or lace shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects, as specified in 4.3.1 through 4.4.3.3, shall not exceed the applicable acceptable quality levels as specified in the contract or purchase order.

3.13 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished lace shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.7.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1).
b. Conformance inspection (see 4.3.2).

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4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects listed in 4.4.3.1 and 4.4.3.2 and tested as specified in 4.4.4. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.3.2 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 through 4.5.3 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits as specified in the contract and/or procurement documents, except where otherwise indicated (see 6.2).

4.4 Visual examination for cloth. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with MIL-STD-3064, Type IV.

4.4.1 Roll identification and marking (face/back) examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064, Type IV.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064, Type IV.

4.4.3 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064, Type IV.

4.4.4 Dimensional examination. The materials shall be examined for conformance to the dimensions specified in 3.8.2 and 3.8.3 when measured in accordance with 4.5.

4.5 End item testing. The braid or lace shall be tested for the characteristics listed in Table VII. The methods of testing as specified wherever applicable and as listed in Table VII shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be 7 yards full width of the finished braid or 4 yards of lace for all physical and chemical tests except for the weight, the lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. For the weight characteristic, acceptance shall be based on conformance with the lot average requirement. The sample size shall be in accordance with the following:

Lot size in yards	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VII. End item testing.

Characteristic	Requirement reference	Test method
Braid only		
Construction:		
Weight, oz/gr yd	Table V	4.5.1
No. of carriers	Table V	Visual 1/
Ends per carrier	Table V	Visual 1/
No. of vellum	Table V	Visual 1/
No. of stuffer yarns	Table V	Visual 1/
No. of warp ends	Table V	Visual 1/
No. of lines	Table V	Visual 1/
Picks per inch	Table V	ASTM-D3775
Breaking strength, warp	Table V	ASTM-D5035 2/
Type of braid	Table V	Visual 1/
Lace only		
Construction:		
Total ends	Table VI	Visual 1/
Weight/lin. yd	Table VI	4.5.1
Picks per inch	Table VI	ASTM-D3775
No. of vellum	Table VI	Visual 1/
Weave	Table VI	Visual 1/
Ribs per inch	Table VI	ASTM-D3775
Breaking strength, warp	Table VI	ASTM-D5035 3/
Dimensional change:		
To dry-cleaning (1 cycle)	3.8.2	AATCC TM158 4/
To laundering (3 cycles)	3.8.3	AATCC TM135 5/
Braid and lace		
Labile sulfur	Free	4.5.2
Colorfastness:		
Fastness to perspiration	3.5.4	AATCC TM15
Fastness to wet dry cleaning	3.5.4	AATCC TM132
Fastness to light	3.5.4	AATCC TM16.3 Opt 1 or 3
Fastness to dry heat (for polyester only)	3.5.4	AATCC TM117 6/
Fastness to crocking	3.5.4	AATCC TM117
Toxicity	3.13	4.5.3

1/ One determination per sample unit and results reported as “actual count”.

2/ Except the machine shall have smooth grips of the drum type not less than 2-inches in diameter. The distance between the center of the drums at the beginning of the tests shall be 6-inches.

3/ Except that the breaking strength of the lace shall be determined by breaking the full width of the lace.

4/ Except the specimen shall be 20-inches long and full width.

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5/ Machine cycle, permanent press; wash temperature, 120ºF (± 5), tumble dry, permanent press.
6/ The test shall be conducted at 376ºF.

4.6 Methods of inspection.

4.6.1 Weight. Select two (2) specimens of the finished item in excess of 1 yard each from each sample unit. Measure the specimens of braid or lace in the relaxed state and cut exactly 1 yard each. While in a standard condition as defined in the specified method, each specimen shall be weighed on a balance or scale accurate to 0.01 ounce (a scale or balance graduated in grams may be used and the results converted to ounces). Average the weight of all braid specimens and multiply this value by 144 to determine ounces per gross yards to the nearest 0.5 ounce; the weight of all lace specimens shall be averaged to determine ounces per linear yard to the nearest 0.01 ounce.

4.6.2 Visual shade matching (all Types, Classes and Styles). The color and appearance of the braid and lace shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.6) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the braid and lace shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.6.1). The braid and lace shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

4.6.3 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g $SnCl_2$ (ACS grade) in 100 mL of 35 percent hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail.

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4.7 Toxicity test. When the toxicity requirement (see 3.13) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2 and 6.7).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or withing the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The braids and laces are used for decorative accessories to the dress uniforms for officers and enlisted personnel of the military. The types, classes and styles are used as follows:

Type I – Enlisted personnel

Class 1 - Coat sleeves
Class 2 - Coat shoulder loops
Class 3 - Blue dress cap
Class 4 - Belt and trouser stripes

Type II – Officers and enlisted personnel

Class 1, Style A - General Officers' trouser stripes and enlisted men's service stripes
Class 1, Style B - Officers' and enlisted men's trouser stripes and General Officers' coat sleeve ornamentation
Class 1, Style C - General Officers' coat sleeve ornamentation
Class 1, Style D - Officers' and enlisted women's slack stripes and General Officers' coat sleeve ornamentation
Class 2, Style A - Officers' coat sleeve ornamentation
Class 2, Style B - Officers' cap band

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and style required (see 1.2).
c. The specific issue of individual documents referenced (see 2.1).
d. When a first article is required (see 3.1, 4.3.1, and 6.3).
e. Colorfastness required (see 3.5.4).
f. When method of put-up is other than specified (see 3.9).
g. When toxicity is required (see 3.13 and 4.7).
h. Packaging requirements (see 5.1).

6.3 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for that specified requirement.

6.6 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.

6.6.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75. DD1222's should reflect which light source is used. If the light source is not indicated on the DD1222, D65 will be assumed by the Government Shade Laboratory.

6.7 Toxicity testing. If dermal toxicity testing is required, it will be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1)

Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100)

Dermal Toxicity (https://law.cornell.edu/cfr/text/40/798.2250)

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6.8 Subject term (key word) listing.

Accoutrements
Decorative lace
Ornamentation

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army – GL

Preparing activity:
DLA-CT

Review activities:
Air Force – 11
Navy – NU

(Project 8315-2025-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.

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